The documentation and process conversion measures necessary to comply with this amendment shall be completed by 28 September 2002.

INCH POUND
MIL-PRF-19500/116L
AMENDMENT 2
28 June 2002
AMENDMENT 1
2 November 2001

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, SWITCHING
TYPES 1N914, 1N914UR, 1N4148-1, 1N4148UR-1, 1N4148UB, 1N4148UB2, 1N4148UB2R, 1N4148UBCA, 1N4148UBCC, 1N4148UBCD, 1N4531, AND 1N4531UR, JAN, JANTX, JANTXV, JANHC, AND JANKC

This amendment forms a part of MIL-PRF-19500/116L, dated 13 July 2001 and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 8

3.7, delete and substitute, "3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.".

PAGE 10

*4.3.1, delete and substitute: "4.3.1 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500, "Discrete Semiconductor Die/Chip Lot Acceptance". Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

"4.3.2 Burn-in test conditions. Burn-in conditions are as follows:

| Type | $T_A$ = room ambient as defined in MIL-STD-750 $V_{RWM}$ = 75 V (pk), f = 50 - 60 Hz | $T_A$ = room ambient as defined in MIL-STD-750 |
|---|---|---|
| 1N914, 1N914UR | $I_O$ = 75 mA | $I_F$ = 150 mA min |
| 1N4531, !N4531UR | $I_O$ = 125 mA | $I_F$ = 175 mA min |
| 1N4148-1, 1N4148UR-1, 1N4148UB, 1N4148UB2, 1N4148UBR2 | $I_O$ = 200 mA | $I_F$ = 200 mA min |

"

(This reverses a change made by amendment 1.)

4.4.2.1, B3, conditions, delete "(see 4.5.1)" and substitute "(see 4.5.2)".

4.4.2.1, B3, conditions, delete "1N914: $I_O$ = 5 mA" and substitute "1N914: $I_O$ = 75 mA".

4.4.2.1, B5, conditions, delete, "$R_{\theta JL}$ 100°C/W (UR)" and substitute "$R_{\theta JC}$ = 100°C/W (UR)".

4.4.3, C6, conditions, delete "(see 4.5.1)" and substitute "(see 4.5.2)".

After 4.4.3, add,

"4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table IX of MIL-PRF-19500, and table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2, herein."


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:15Z
Check the source to verify that this is the current version before use.

PAGE 11

4.5.6, third sentence, after "$R_{\theta JL}$" add: "and $R_{\theta JEC}$".

PAGE 13

*TABLE I, Inspection heading, add "2/".

*TABLE I, subgroup 2, reverse current, delete "1N914, 1N4531, 1N4148-1".

TABLE I, subgroup 2, reverse current, symbol $I_{R1}$, min limits column, delete "25" three places and substitute in max limits column "25" three places.

TABLE I, subgroup 2, reverse current, symbol $I_{R2}$, min limits column, delete "500" and substitute in max limits column "500".

*TABLE I, subgroup 2, forward voltage, inspection column, add "1N914" for the $I_F$ = 50 mA dc.

*TABLE I, subgroup 2, forward voltage, inspection column, add "For all other devices" for the $I_F$ = 100 mA dc.

TABLE I, subgroup 2, forward voltage, symbol $V_{F2}$, min limits column, delete "1.2" three places and substitute in max limits column "1.2" two places.

*TABLE I, subgroup 3, reverse current, $I_{R3}$ delete "1N914, 1N4531, 1N4148-1".

*TABLE I, subgroup 3, reverse current, $I_{R4}$ delete "1N914, 1N4531, 1N4148-1".

*TABLE I, subgroup 3, reverse current, $I_{R4}$ limits, delete "75" two places.

*TABLE I, subgroup 3, reverse current, $I_{R4}$ units, delete "µA dc" two places.

*TABLE I, subgroup 3, reverse current, $V_{F3}$ delete "1N914, 1N4531, 1N4148-1".

*TABLE I, subgroup 3, reverse current, $V_{F3}$ delete "0.8" two places.

*TABLE I, subgroup 3, reverse current, $V_{F3}$ units, delete "V dc" two places.

PAGE 14

*TABLE I, Inspection heading, add "2/".

*TABLE I, subgroup 3, forward voltage, delete "1N914, 1N4531, 1N4148-1".

*TABLE I, subgroup 3, forward voltage, inspection column, add "1N914" for the $I_F$ = 50 mA dc.

*TABLE I, subgroup 3, forward voltage, inspection column, add "For all other devices" for the $I_F$ = 100 mA dc.

*TABLE I, subgroup 4, junction capacitance, delete "1N914, 1N4531, 1N4148-1".

*TABLE I, subgroup 4, junction capacitance, limits, delete "4.0" two places.

*TABLE I, subgroup 4, junction capacitance, units, delete "pF" two places.

*TABLE I, subgroup 4, reverse recovery time, delete "1N914, 1N4531, 1N4148-1"



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:15Z
Check the source to verify that this is the current version before use.

PAGE 15

*TABLE I, Inspection heading, add “2/”.

TABLE I, subgroup 7, delete “(see 4.5.2)” and substitute “(see 4.5.3)”.

*TABLE I, end of table, add “2/ UBCA, UBCC, and UBCD devices are to have each diode tested individually.”

PAGE 16

TABLE II, subgroup 4, thermal resistance leaded devices, delete “$R_{\theta JEC}$” and substitute “$R_{\theta JL}$”.

PAGE 17

6.4, delete “(MIL-S-19500/578)” and substitute “(MIL-PRF-19500/578)”.

PAGE 18

6.5, delete “QPL” and substitute “QML”.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
Air Force - 19

Preparing activity
DLA - CC

(Project 5961-2602)



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:15Z
Check the source to verify that this is the current version before use.